

Topside sounder retrieval

TOPSIDE SOUNDER RETRIEVAL | 25 SEPTEMBER 2026

Status, recovery evidence, and truth-independence audit

Executive summary

The retrieval code can generate and fit synthetic vertical and oblique topside ionograms. This report evaluates one reduced vertical case and asks whether a low ionogram mismatch also means that the electron-density profile was recovered. No external observations are used.

Principal findings

Closed-model recovery

A true density scale of 1.12 was recovered as 1.00 by a six-point grid search (absolute error 0.12).

Profile-shape mismatch

A separately specified altitude perturbation produced a 6.7% profile NRMSE over 150–450 km, despite a small ionogram objective.

No-return failure

An empty truth ionogram and an empty candidate scored zero objective while their profiles differed by 32.0% NRMSE.

Validation verdict

The current demo's truth is generated by the same density parameterization, PyIRI background, ray tracer, and image formation used in retrieval. The alternate profile test changes only the density formula. Fully independent truth has not been established.

Read pages 2–6 for setup, paired ionograms, interpretation, and the validation gate.

1. Methods and provenance

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One generated 800 km Earth orbit supplies a vertical transmitter/receiver case. The background is PyIRI; the scale search uses 4, 5, and 6 MHz. Only density scale is searched, at 0.80 through 1.30 in 0.10 steps. The displayed ionograms are re-rendered from 2 to 10 MHz every 100 kHz in 1 km range bins. Homing tolerance is 1,000 m, with up to 10 accepted returns per frequency and mode. Every accepted O/X return is plotted from 0 km, including short-range returns below the 150 km retrieval floor. Returns sharing a bin are summed; no smoothing is applied. The retrieval objective uses 150–2,600 km. This is a grid-search diagnostic, not the full five-parameter optimizer.

Where each truth comes from

Case	Density truth	PyIRI	PHaRLAP	External data
Closed model	Fitted scalar family	Shared	Shared	No
Shape mismatch	Separate altitude formula	Shared	Shared	No
No-return control	Separate altitude formula	Shared	Shared	No

Table 1. Truth provenance. “Shared” means the truth generator and candidate model use the same component.

Scoring and accuracy measures

The ionogram objective combines normalized total-power error, O/X split error, and Doppler error. The profile NRMSE is root-mean-square density error divided by root-mean-square truth density, evaluated at the center grid column from 150 to 450 km. The ionogram objective has no direct percent interpretation.

Independence conclusion

A separate formula is useful for testing shape mismatch, but both sides still share Earth geometry, PyIRI, PHaRLAP, and ionogram construction. It is not a fully independent ground truth.

2. Closed-model ionograms

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The synthetic truth uses density scale 1.12. A search over six candidate scales selected 1.00. Both ionograms below were re-rendered from 2 to 10 MHz every 100 kHz in 1 km range bins; only 4, 5, and 6 MHz selected the parameter.

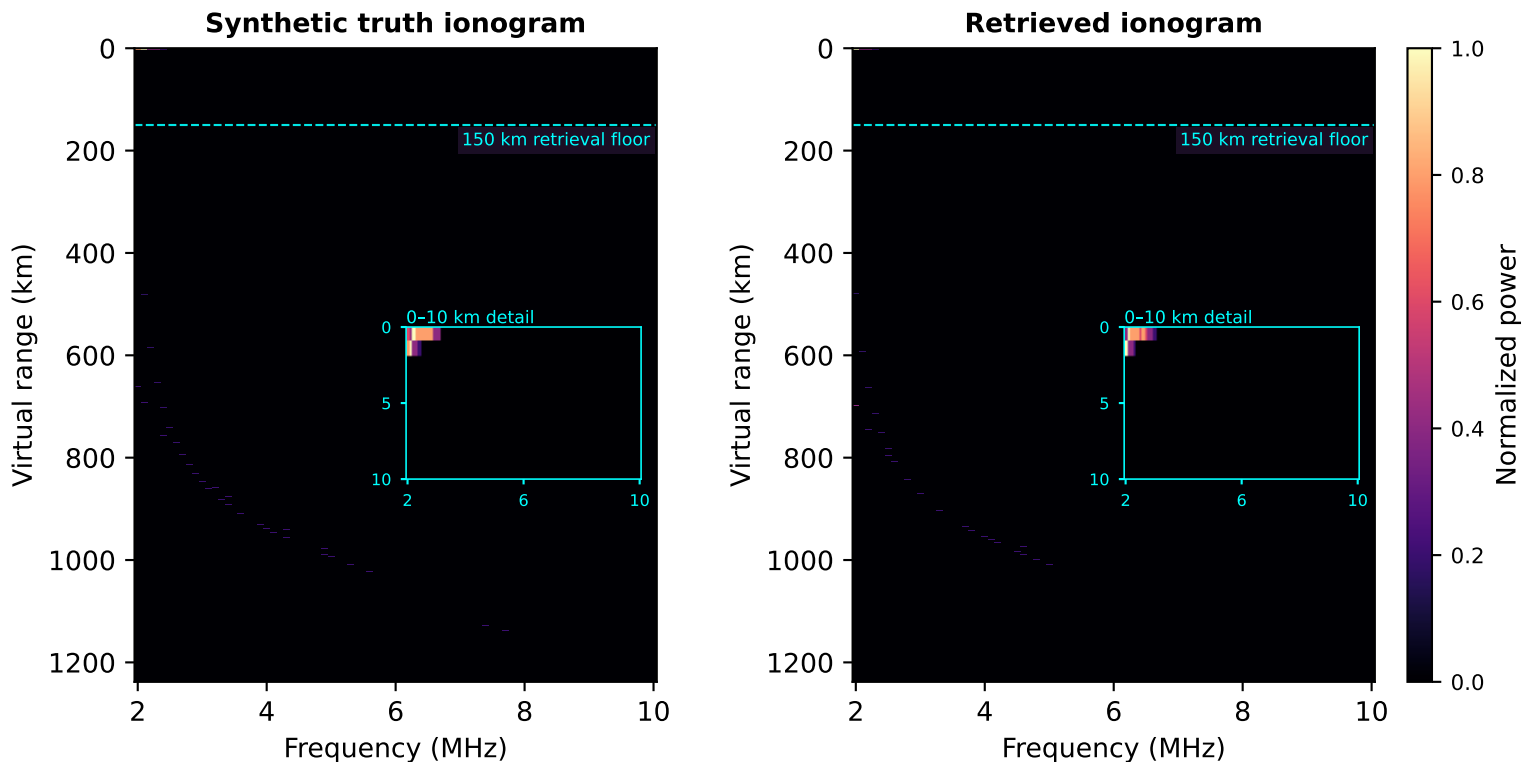


Figure 1. Synthetic truth (left) and retrieved (right) ionograms show every accepted O/X return, including those below the dashed 150 km retrieval floor. Insets enlarge 0–10 km. Returns in the same 1 km bin are summed, with no smoothing. Axes and color limits match; each image is normalized to its own peak, so absolute received power cannot be compared.

What to look for

Compare the frequency and virtual-range locations of the bright returns. Of 85 accepted truth returns, 54 lie below 10 km and are outside the score. There are 21 frequency/mode cells with multiple accepted returns. The 1,000 m homing gate checks receiver miss distance; it does not suppress multipath. The parameter error and search objective appear on page 4.

3. Closed-model recovery

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First, the truth density is generated by the same scalar family searched by the retrieval. The generating scale (1.12) is deliberately absent from the six candidates.

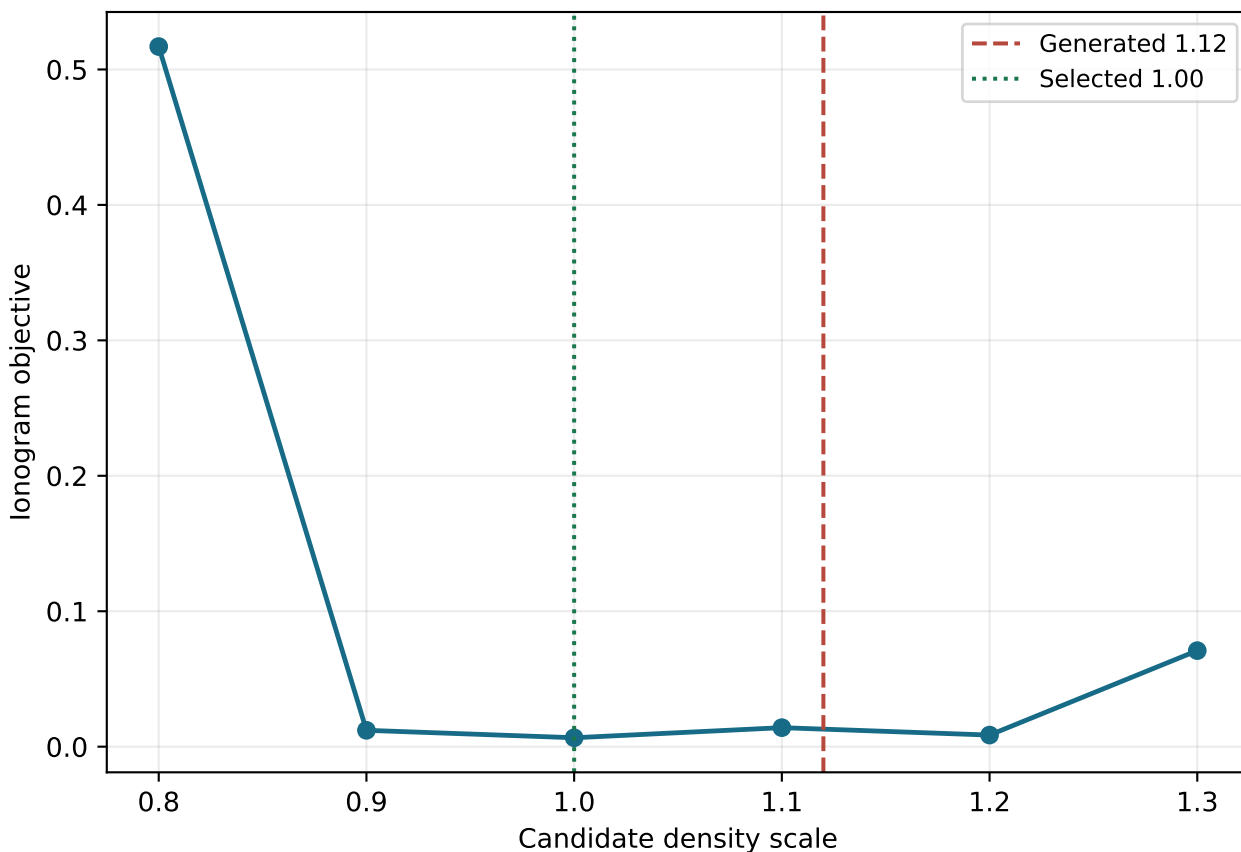


Figure 2. The six-point search selected scale 1.00 for truth 1.12. The objective is not smooth: neighboring candidates can change the available ray returns. This plot shows the complete set of evaluated candidates, not a continuous fit curve.

Interpretation

The absolute scale error is 0.12 (10.7% of truth). Because the generator and fitter share the same physical and numerical assumptions, this is a check of recoverability in one favorable model family, not independent validation.

4. Shape mismatch: profile and ionograms

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Here truth uses an altitude-dependent density factor, $1.10 + 0.10 \exp[-0.5((h - 320 \text{ km})/55 \text{ km})^2]$. Candidates can only multiply the PyIRI background by a constant. The search selected scale 1.10.

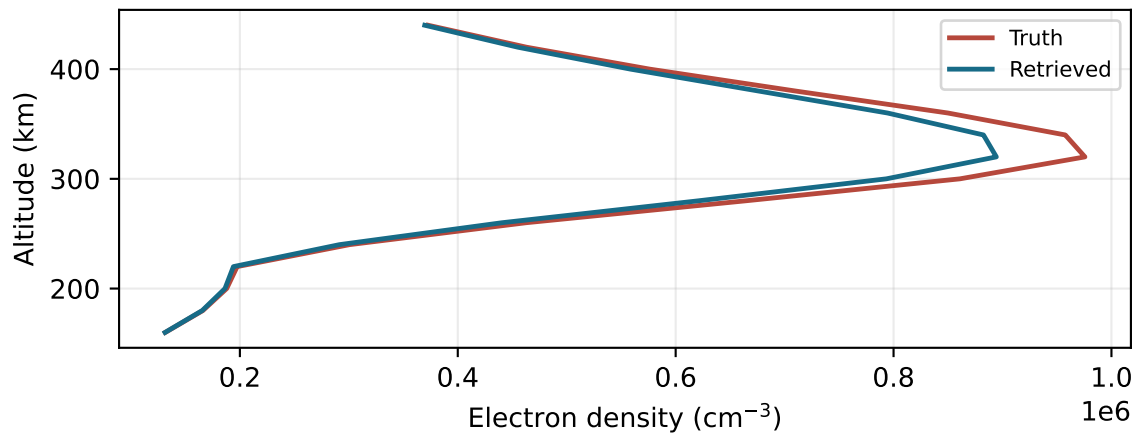


Figure 3. Truth and retrieved density differ by 6.7% NRMSE over 150–450 km at the center grid column.

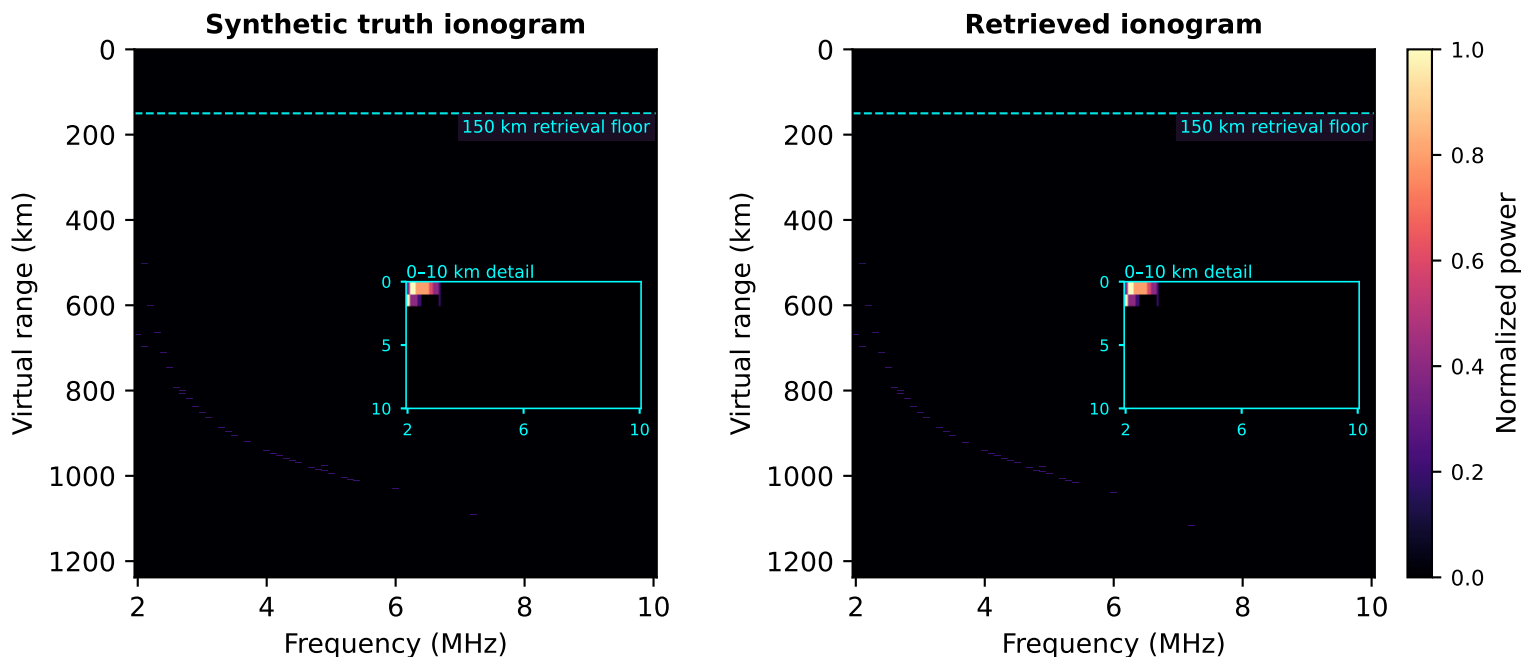


Figure 4. Truth (left) and retrieved (right) ionograms from 2 to 10 MHz at $100 \text{ kHz} \times 1 \text{ km}$ show all accepted O/X returns, including those below the dashed 150 km retrieval floor; insets enlarge 0–10 km. Coincident returns are summed without smoothing. Each panel is normalized to its own peak. The 4, 5, and 6 MHz search objective is 0.00014834, yet the vertical profile still differs by 6.7%. The truth formula is separate; the background and ray tracer are shared.

5. No-return failure and next gate

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A second altitude-dependent truth gives no returns at 4–6 MHz. The scale-0.80 candidate also gives no returns, so both ionograms are empty and their objective is zero.

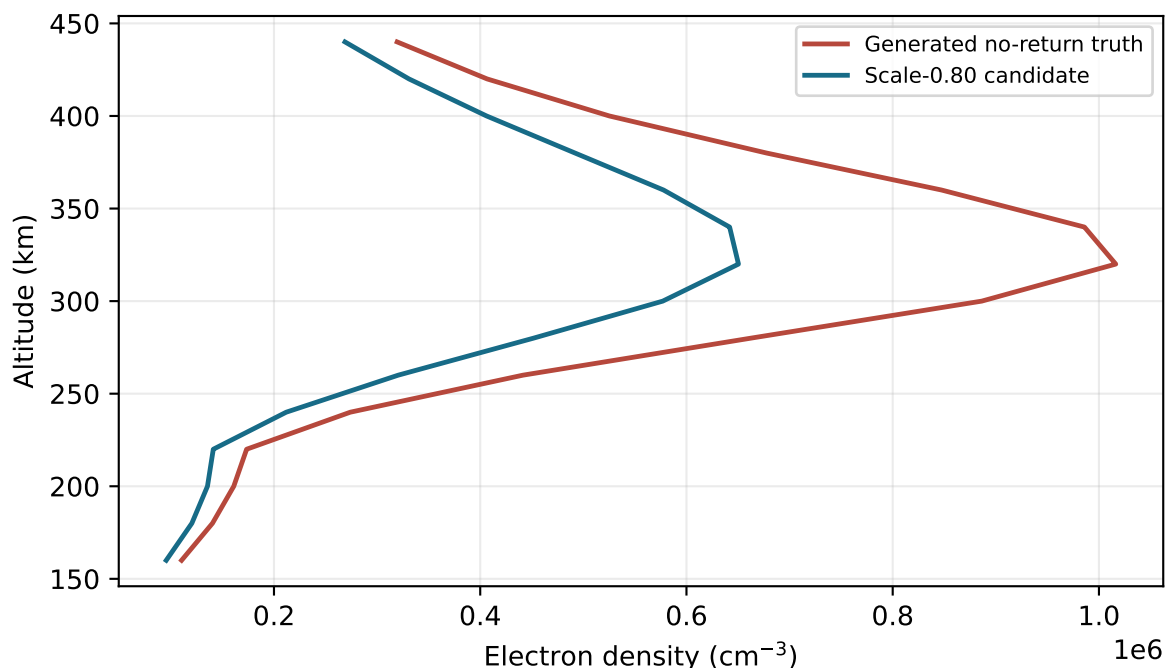


Figure 5. Profiles differ by 32.0% NRMSE from 150 to 450 km while the ionogram objective is exactly zero. The score cannot distinguish two empty observations. Such cases must be flagged and excluded from accuracy claims.

Conclusion

The present evidence supports a working synthetic diagnostic and exposes weak identifiability. It does not confirm independent truth or measured retrieval accuracy.

Required validation

Use an external density field or measured sounder profiles with matching geometry. Keep the truth source outside the retrieval forward model, include frequencies with detected returns, and report density bias and NRMSE across multiple cases.